# SWEKSHA TUDU

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## SUMMARY

AI/ML Engineer with 1+ year of experience designing, deploying, and operating production Generative AI and machine learning systems for enterprise clients, spanning LLM applications, retrieval-augmented generation (RAG), LLM evaluation and guardrails, predictive modeling, and MLOps. Hands-on with Python, SQL, LangChain, LangGraph, and LLM provider APIs, deployed on AWS and Azure. AWS Certified Cloud Practitioner.

## EXPERIENCE

**Grid Dynamics**

Aug. 2025 - Jul. 2026

Junior Data Scientist

Bangalore, India

**RAGaaS: RAG as a Service**

- Built and scaled a multi-tenant RAG platform serving 90+ enterprise clients, 10K+ daily users, and 100K+ documents, with decoupled ingestion and query APIs and configuration-driven onboarding requiring zero code deployments.
- Designed hybrid keyword + semantic retrieval with dual-index ACL filtering, reranking, and query-rewrite fallbacks, improving Precision@5 from 78% to 91% and Recall@5 from 72% to 89% across 100K+ enterprise documents, with citation-backed responses streamed to users.

**Model Service: LLM Gateway & Platform Pipelines**

- Built an internal LLM gateway in Python (Flask, FastAPI) exposing one OpenAI-format API across OpenAI, AWS Bedrock, Vertex AI, and Azure, with tool-calling format translation, silent failover, PII masking, and quota checks before vendor calls.
- Added reliability controls: atomic Redis (Lua) rate limiting and cost tracking, idempotent batch processing, and LangSmith tracing; onboarded 20+ model versions via a config-driven registry and cut integration-test time ~55% (18 to 8 min).
- Orchestrated configuration releases across 30+ production servers using controlled restart workflows, REST API validation, and per-server health checks, delivering zero-defect deployments and continuous service availability for systems supporting 10K+ daily users.

**Grid Dynamics**

Jan. 2025 - Jul. 2025

Data Science Intern

Hyderabad, India

**Red Teaming System for AI Agents: Evaluation & Guardrails**

- Built an evaluation and guardrail framework (LangChain, ChromaDB) assessing agentic AI/LLM robustness in a sandboxed environment with real-time monitoring, identifying failure modes ahead of production deployment.
- Designed structured test scenarios simulating prompt-injection and multi-turn attacks; performed data-driven analysis to quantify failure rates and deliver findings that informed model-hardening decisions.

**Developer Profiling via AST & Time Series Modeling: Predictive Modeling**

- Designed and built predictive ML models on large-scale repository data: a churn classifier achieving 70% accuracy (beating the majority-class baseline) and a regression model estimating experience level, generating actionable insights for stakeholders.
- Engineered temporal and statistical features from complex raw data using Tree-sitter AST analysis; applied SHAP/LIME for model interpretability and built Tableau/Power BI dashboards to communicate results.

## PROJECT

**JARVISTECHIES: AI Recruitment Platform Hackathon**

- Built an end-to-end GenAI pipeline computing semantic similarity via embeddings and integrating Whisper transcription with GPT-driven follow-up question generation for a recruitment app.
- Engineered a Resume Analyzer API with skills-based and semantic scoring, delivered via a unified API layer powering a dynamic frontend within a sandboxed, monitored architecture.

## EDUCATION

**Birla Institute of Technology, Mesra**

2021 - 2025

B.Tech, Electronics & Communication Engineering

Ranchi, India

## TECHNICAL SKILLS

Programming: Python, SQL, REST APIs

Generative AI & NLP: LLMs, RAG, LangChain, LangGraph, Agentic Workflows, NLP, Prompt Engineering, Embeddings

LLM Evaluation: Guardrails, Evaluation Frameworks, Adversarial/Red-Team Testing

ML & Statistics: Machine Learning, Classification, Regression, Predictive Modeling, Feature Engineering, SHAP/LIME

MLOps & Production: Model Deployment, ML Pipelines, MLOps, Docker, FastAPI, CI/CD, Grafana, Splunk, Prometheus

Cloud: AWS (Certified Cloud Practitioner), Azure (OpenAI, App Configuration)

Big Data & Processing: Apache Spark, Redis, pandas, NumPy, batch & streaming data pipelines

Databases & Visualization: PostgreSQL, MySQL, ChromaDB, JSON, Tableau, Power BI, matplotlib